COMSATS University Islamabad, Vehari Campus,Pakistan

Department of Computer Science

Ghost Interview (AI Mock Interviewer)



Software Requirements Speci cation (SRS)

Bachelor of Science in Computer Science

Submitted By

Ehsan Hamza SP23-BCS-019

Shaharyar Altaf SP23-BCS-055

Supervisor

Sir Qizar Javed

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Chapter 3

Requirement Analysis

3.1 Introduction to Requirement Analysis

This chapter presents a structured and formal speci cation of system requirements for Ghost Interview AI Mock Interviewer. The purpose of this chapter is to clearly de ne what the system shall accomplish before proceeding to the system design phase. The requirements documented here serve as a contractual foundation for implementation, validation, and system testing.

The requirement analysis includes requirement elicitation techniques, identi cation of stakeholders, use case modeling, functional requirements, non-functional requirements, and external interface requirements. All requirements are written in measurable and testable format using standard requirement engineering principles.

3.2 Requirement Elicitation Techniques

Requirements for this project were identi ed using the following techniques:

Stakeholder Interviews: To capture user expectations and constraints from students, job seekers, and supervisors.

Literature Review: Analysis of research papers on emotion detection, speech analysis, and interview preparation systems.

Analysis of Existing Solutions: Examination of HireVue, Pramp, and Interviewing.io to identify functional gaps.

Online Questionnaires: To understand user requirements and pain points regarding interview preparation.

AI Documentation Review: Study of OpenAI Whisper, Gemini API, DeepFace, and ElevenLabs documentation.

3.3 User Classes and Characteristics

This section identi es di erent user categories interacting with the system. Each user class is de ned along with their responsibilities, access privileges, and technical pro ciency level.

Table 3.1: User Classes and Characteristics

User Class	Description	Technical Level
Administrator	Manages system con guration, user accounts, monitoring, and reports.	Advanced
End User (Candidate)	Uses the system to select job roles, conduct AI mock interviews, receive performance reports, and track improvement.	Basic
External System	AI APIs and third-party services (Whisper, Gemini, DeepFace, ElevenLabs) interacting with the application backend.	Technical

3.4 System Overview

The proposed system operates within a three-tier web architecture designed to support modular interaction between user interfaces, backend processing logic, AI services, and data storage. The system supports role-based interactions and ensures secure communication between internal modules and external AI services.

The system boundary includes:

React.js-based frontend (user interface and webcam/microphone interaction).

Node.js backend (business logic, API orchestration, session management).

AI processing layer (speech-to-text, emotion detection, language model, voice synthesis).

Database (user data, interview sessions, reports).

3.5 Use Case Model

The use case model represents high-level system interactions between actors and system functionalities. It de nes system boundaries and captures functional behavior from a user perspective.

3.5.1 Use Case Diagram

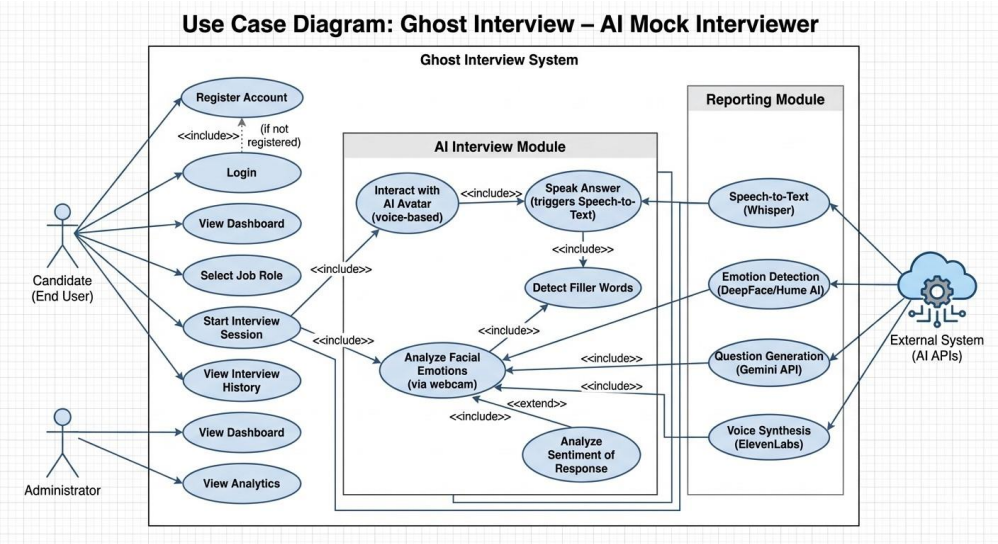


Figure 3.1: Use Case Diagram of Ghost Interview System

The diagram illustrates the interaction between the Candidate, Administrator, and external AI systems with the major system functionalities including login, job role selection, interview session, speech analysis, emotion analysis, and report generation.

3.5.2 Use Case Descriptions

Module 1: User Management

Module 2: AI Interview Module

Module 3: Reporting and Dashboard

3.5.3 Event-Response Table

3.6 Functional Requirements

Functional requirements de ne the core services and behaviors that the system shall provide. These requirements are derived directly from the use case model. Each requirement is uniquely identi ed, measurable, and written using formal speci cation language following IEEE-style structure, categorized by system modules.

3.6.1 Module 1: User Management

FR-1.1 User Registration

Table 3.2: UC-1.1: User Registration

Use Case ID	UC-1.1
Use Case Name	User Registration
Actors	End User (Candidate)
Description	This use case describes how a new user registers into the Ghost Interview system.
Trigger	User selects the Sign Up option on the homepage.
Preconditions	User is not already registered.
Postconditions	User account is successfully created and stored in the database.
Normal Flow	1. User navigates to the registration page. 2. User enters name, email, and password. 3. System validates input elds. 4. System stores data securely in the database. 5. Con rmation email is sent and success message isdisplayed.
Alternative Flow	If validation fails, the system displays appropriate error messages and prompts the user to correct the input.

Table 3.3: UC-1.2: User Login

Use Case ID	UC-1.2
Use Case Name	User Login
Actors	End User, Administrator
Description	This use case describes how a registered user logs into the system.
Trigger	User selects the Login option.
Preconditions	User has a registered account.
Postconditions	User is authenticated and redirected to the dashboard.
Normal Flow	1. User enters email and password. 2. System validates credentials against stored records. 3. System grants access and redirects to the dashboard.
Alternative Flow	If credentials are invalid, the system displays an error message and prompts re-entry.

FR-1.2 User Authentication

Table 3.4: UC-2.1: Job Role Selection and Interview Initiation

Use Case ID	UC-2.1
Use Case Name	Job Role Selection and Interview Initiation
Actors	End User
Description	The user selects a job role and initiates an AI-driven mock interview session.

Trigger	User clicks Start Interview after selecting a job role.
Preconditions	User is logged in; webcam and microphone are accessible.
Postconditions	Interview session is started and AI avatar begins asking questions.
Normal Flow	1. User selects a job role from the available list. 2. System loads relevant interview questions via Gemini API. 3. AI avatar voice is activated through ElevenLabs. 4. Avatar asks the rst interview question. 5. Webcam and microphone recording begins.
Alternative Flow	If webcam or microphone access is denied, the system displays a permission error and halts the session.

Table 3.5: UC-2.2: Speech Analysis

Use Case ID	UC-2.2
Use Case Name	Speech Analysis
Actors	End User, External System (Whisper API)
Description	The system converts the user's spoken response to text and identi es ller words.
Trigger	User speaks in response to the AI avatar's question.
Preconditions	Interview session is active; microphone access is granted.
Postconditions	Text transcript and ller word count are stored for the session.
Normal Flow	1. User speaks their answer. 2. Audio is captured and sent to OpenAI WhisperAPI. 3. Whisper converts audio to text. 4. System scans transcript for ller words (e.g., um ,uh). 5. Results are stored in the session data.
Alternative Flow	If audio quality is too low, system ags low-quality recording and requests user to repeat.

3.6.2 Module 2: AI Interview Module

FR-2.1 Job Role Selection

Table 3.6: UC-2.3: Emotion and Sentiment Analysis

Use Case ID	UC-2.3
Use Case Name	Emotion and Sentiment Analysis
Actors	End User, External System (DeepFace / Hume AI)
Description	The system analyzes the user's facial expressions and emotional tone during the interview.

Trigger	Video frames are captured during the interview session.
Preconditions	Webcam access is granted; interview session is active.
Postconditions	Emotion metrics (con dence, nervousness, eye contact) are stored.
Normal Flow	1. Webcam frames are captured at regular intervals. 2. Frames are sent to DeepFace / Hume AI for analysis. 3. Emotion labels and con dence scores are returned. 4. Sentiment of text transcript is analyzed. 5. All metrics are stored in the session database.
Alternative Flow	If camera feed is unavailable, emotion analysis is skipped and user is noti ed.

Table 3.7: UC-3.1: Con dence and Competency Report Generation

Use Case ID	UC-3.1
Use Case Name	Con dence and Competency Report Generation
Actors	End User
Description	After the interview session, the system generates a detailed performance report covering speech, emotion, and competency metrics.
Trigger	Interview session is completed.
Preconditions	At least one interview session has been completed successfully.
Postconditions	Report is generated, stored, and displayed to the user.
Normal Flow	1. System aggregates speech, emotion, and sentimentdata. 2. System computes con dence score, ller word frequency, and emotion timeline. 3. Report is formatted and stored in the database. 4. User is redirected to the Report screen with insights.
Alternative Flow	If session data is incomplete, system generates a partial report and noti es the user.

FR-2.2 AI Avatar Interview Conduction

FR-2.3 Speech-to-Text Conversion

Table 3.8: Event-Response Table

Event	Source	System Response

User Login Attempt	End User	Validate credentials and grant or deny access.
Job Role Selection	End User	Load interview questions from Gemini API.
Interview Started	End User	Activate AI avatar, begin webcam and microphone recording.
User Speaks Answer	End User	Capture audio and send to Whisper for transcription.
Video Frame Captured	System	Send frame to emotion detection model; store result.
Interview Completed	System	Aggregate data and generate Con dence & Competency Report.
Report Requested	End User	Retrieve stored report and display on dashboard.
API Unavailability	External System	Display error message and pause session gracefully.

Table 3.9: Functional Requirement User Registration

Identi er	FR-1.1
Title	User Registration
Requirement	The system shall allow users to create an account by providing a valid name, email address, and password.
Source	Stakeholder Requirement
Rationale	Enables authorized access to system services and personalized interview tracking.
Dependencies	Database Connectivity
Priority	High

Table 3.10: Functional Requirement User Authentication

Identi er	FR-1.2
Title	User Authentication
Requirement	The system shall authenticate users using email and password and maintain secure sessions.
Source	Stakeholder Requirement
Rationale	Ensures only registered users can access interview and reporting features.
Dependencies	FR-1.1 (User Registration), Database Connectivity
Priority	High

FR-2.4 Filler Word Detection

Table 3.11: Functional Requirement Job Role Selection

Identi er	FR-2.1
Title	Job Role Selection

Requirement	The system shall allow users to select from prede ned job roles and interview categories before initiating a session.
Source	System Requirement Speci cation
Rationale	Enables domain-speci c question generation tailored to the user's target role.
Dependencies	FR-1.2, Gemini API Integration
Priority	High

Table 3.12: Functional Requirement AI Avatar Interview

Identi er	FR-2.2
Title	AI Avatar Interview Conduction
Requirement	The system shall conduct voice-based interviews through an AI avatar that asks contextually relevant questions using a realistic human-like voice via the ElevenLabs API.
Source	System Requirement Speci cation
Rationale	Provides a realistic and immersive mock interview experience.
Dependencies	FR-2.1, Gemini API, ElevenLabs API, WebRTC
Priority	High

Table 3.13: Functional Requirement Speech-to-Text Conversion

Identi er	FR-2.3
Title	Speech-to-Text Conversion
Requirement	The system shall convert the user's spoken responses into text using OpenAI Whisper and store the transcript for analysis.
Source	System Requirement Speci cation
Rationale	Enables verbal response evaluation and ller word detection.
Dependencies	OpenAI Whisper API, Microphone Access via WebRTC
Priority	High

FR-2.5 Facial Emotion Detection

FR-2.6 Sentiment Analysis

Table 3.14: Functional Requirement Filler Word Detection

Identi er	FR-2.4
Title	Filler Word Detection

Requirement	The system shall identify and count ller words (e.g., um, uh, like, you know) in the speech transcript.
Source	System Requirement Speci cation
Rationale	Provides actionable feedback on speech quality and uency.
Dependencies	FR-2.3 (Speech-to-Text)
Priority	Medium

Table 3.15: Functional Requirement Facial Emotion Detection

Identi er	FR-2.5
Title	Facial Emotion Detection
Requirement	The system shall analyze webcam video frames using DeepFace or Hume AI to detect facial emotions, eye contact, and con dence indicators during the interview.
Source	System Requirement Speci cation
Rationale	Provides non-verbal behavior analysis, a key component of comprehensive interview assessment.
Dependencies	Webcam Access via WebRTC, DeepFace / Hume AI API
Priority	High

Table 3.16: Functional Requirement Sentiment Analysis

Identi er	FR-2.6
Title	Sentiment Analysis
Requirement	The system shall evaluate the emotional tone (positive, negative, neutral) of the user's transcribed responses using NLP-based sentiment analysis.
Source	System Requirement Speci cation
Rationale	Complements emotion detection with language-level emotional assessment.
Dependencies	FR-2.3 (Speech-to-Text)
Priority	Medium

Table 3.17: Functional Requirement Report Generation

Identi er	FR-3.1
Title	Con dence and Competency Report Generation
Requirement	The system shall generate a detailed Con dence and Competency Report after each interview session, including speech quality, emotion timeline, ller word count, sentiment summary, and an overall readiness score.
Source	Business Requirement

Rationale	Supports user self-assessment and performance improvement.
Dependencies	FR-2.3, FR-2.4, FR-2.5, FR-2.6, Database Module
Priority	High

3.6.3 Module 3: Reporting and Analytics

FR-3.1 Con dence and Competency Report Generation

FR-3.2 Interview History Dashboard

Table 3.18: Functional Requirement Interview History Dashboard

Identi er	FR-3.2
Title	Interview History Dashboard
Requirement	The system shall provide users with a dashboard displaying their interview history, past reports, performance trends, and analytics.
Source	Business Requirement
Rationale	Enables users to track improvement over time and review past sessions.
Dependencies	FR-3.1, Database Module, FR-1.2
Priority	Medium

3.7 Non-Functional Requirements

Non-functional requirements de ne quality attributes that describe how the system performs. These requirements are measurable and support system reliability, usability, performance, security, and maintainability.

3.7.1 Performance Requirements

PER-1: The system shall load the main dashboard within 3 seconds under normal operating conditions.

PER-2: Speech-to-text transcription responses from OpenAI Whisper shall be returned within 5 seconds of audio submission.

PER-3: Emotion detection results shall be processed within 2 seconds per video frame batch.

PER-4: The system shall handle at least 50 concurrent interview sessions without signi cant degradation in response time.

PER-5: The AI avatar's rst question shall be delivered within 3 seconds of the interview session start.

3.7.2 Security Requirements

SEC-1: The system shall implement secure user authentication using hashed passwords (e.g., bcrypt).

SEC-2: All data transmission shall use HTTPS protocol to ensure encryption in transit.

SEC-3: Webcam and microphone access shall require explicit user permission via browser consent prompts.

SEC-4: Role-based access control shall be enforced (Admin vs Candidate).

SEC-5: API keys for third-party services shall be stored server-side and never exposed to the client.

SEC-6: User interview recordings and session data shall be stored securely and not shared with third parties.

3.7.3 Reliability Requirements

REL-1: The system shall maintain at least 99% uptime during the evaluation period.

REL-2: System failures shall not result in loss of completed interview session data.

REL-3: The system shall implement error handling for third-party API unavailability and notify users gracefully.

REL-4: Backup and recovery mechanisms shall be implemented for the database.

3.7.4 Usability Requirements

USE-1: The interface shall be intuitive and easy to navigate without prior training.

USE-2: Clear and meaningful error messages shall be displayed for all failure scenarios.

USE-3: The system shall follow standard UI/UX design principles and be visually appealing.

USE-4: The application shall be fully responsive across desktop and laptop screen sizes.

USE-5: The AI avatar interaction interface shall clearly indicate when the avatar is speaking and when the user is expected to respond.

3.7.5 Scalability Requirements

SCA-1: The system architecture shall support increasing number of registered users and concurrent sessions.

SCA-2: The backend shall allow horizontal scaling through cloud deployment.

SCA-3: The database design shall support large volumes of session transcripts and report data.

SCA-4: New job roles and interview categories shall be addable without major restructuring.

3.7.6 Maintainability Requirements

MAIN-1: The system shall follow modular architecture principles with clearly separated frontend, backend, and AI service layers.

MAIN-2: Source code shall be documented clearly using inline comments and README les.

MAIN-3: Future updates, including additional AI models or job roles, shall be integrable without major system restructuring.

MAIN-4: The system shall use version-controlled repositories (e.g., GitHub) for code management.

3.8 External Interface Requirements

This section de nes how the system interacts with external entities including users, hardware devices, software systems, and communication networks.

3.8.1 User Interface Requirements

UI-1: The system shall provide a graphical web-based user interface accessible via modern browsers.

UI-2: Standard font sizes (12 14pt) shall ensure readability throughout the application.

UI-3: Layout shall be responsive across desktop and large-screen devices.

UI-4: Navigation shall remain consistent across all modules (login, role selection, interview, report, dashboard).

UI-5: The interview screen shall display a live webcam feed, AI avatar, and question text simultaneously.

3.8.2 Hardware Interface Requirements

HI-1: The system shall operate on standard computing devices (PC, laptop) with a modern web browser.

HI-2: The system requires access to a functional webcam and microphone for interview session functionality.

HI-3: Minimum hardware: 4GB RAM, modern dual-core processor, HD webcam (720p or higher recommended for accurate emotion detection).

3.8.3 Software Interface Requirements

SI-1: The system shall interact with a relational or non-relational database for storing user data, sessions, and reports.

SI-2: The system shall integrate with the Gemini API for AI-driven interview question generation.

SI-3: The system shall integrate with OpenAI Whisper API for speech-to-text conversion.

SI-4: The system shall integrate with DeepFace / Hume AI for facial emotion detection.

SI-5: The system shall integrate with ElevenLabs API for realistic AI voice synthesis.

SI-6: The frontend shall be built using React.js (v18+) and the backend using Node.js (v20+).

3.8.4 Communication Interface Requirements

CI-1: The system shall use HTTPS protocol for all secure client-server communication.

CI-2: RESTful APIs shall be used for backend communication between frontend and server.

CI-3: WebRTC shall be used for real-time webcam and microphone access within the browser.

CI-4: All third-party API calls shall be made server-side to protect API credentials.

3.9 Assumptions and Dependencies

The system assumes stable internet connectivity for all users during interview sessions.

Third-party services (Gemini, Whisper, DeepFace, ElevenLabs) are assumed to remain available and functional throughout the project.

Users are assumed to possess basic digital literacy and be comfortable using web browsers.

Required cloud hosting infrastructure is assumed to be available during deployment.

Users are assumed to have access to a functional HD webcam and microphone.

Pre-trained AI models will be used; no custom model training is assumed within project scope.

3.10 Requirement Traceability Matrix

This matrix maps project objectives to functional requirements to ensure alignment and completeness.

Table 3.19: Requirement Traceability Matrix

Objective	Use Case	Functional Requirement
Provide user access management	UC-1.1, UC-1.2	FR-1.1, FR-1.2
Conduct AI mock interview sessions	UC-2.1, UC-2.2	FR-2.1, FR-2.2, FR-2.3
Analyze speech quality	UC-2.2	FR-2.3, FR-2.4
Analyze non-verbal behavior	UC-2.3	FR-2.5, FR-2.6
Generate performance reports	UC-3.1	FR-3.1
Provide interview history & analytics	UC-3.1	FR-3.2

Chapter Summary

This chapter presented a structured requirement analysis of the Ghost Interview AI Mock Interviewer system. It de ned requirement elicitation techniques, user classes, use case models, functional requirements, non-functional requirements, and external interface requirements. A traceability matrix was included to ensure alignment between objectives

and implementation requirements. These requirements form the formal foundation for the system design presented in the next chapter.